

$$x = 1, y = 3, z = 3$$

assignment - VII

2 marks

21. If $\text{adj}(A) = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$ then

find A^{-1}

$$|\text{adj} A| = \begin{vmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{vmatrix}$$

$$= 0 + 2(36 - 18) + 0$$

$$= 2(18) = 36$$

$$A^{-1} = \pm \frac{1}{\sqrt{|\text{adj} A|}} (\text{adj} A)$$

$$= \pm \frac{1}{\sqrt{36}} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$$

$$= \pm \frac{1}{6} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 2 & 6 \end{bmatrix}$$

22. Find The Principle argument

Arg Z when $Z = \frac{-2}{1+i\sqrt{3}}$

$$\arg Z = \arg \left(\frac{-2}{1+i\sqrt{3}} \right)$$

$$= \arg(-2) - \arg(1+i\sqrt{3})$$

$$= \pi - \tan^{-1} \left(\frac{0}{2} \right) - \tan^{-1} \left(\frac{\sqrt{3}}{1} \right)$$

$$= \pi - 0 - \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

23. Find The Equation of The hyperbola $(0, \pm 4)$ and Foci $(0, \pm 6)$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 4$$

$$a^2 = 16$$

$$c^2 = a^2 + b^2$$

$$c^2 - a^2 = b^2$$

$$36 - 16 = b^2$$

$$b^2 = 20$$

Eq of hyperbola

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$\frac{y^2}{16} - \frac{x^2}{20} = 1$$

$$5y^2 - 4x^2 = 80$$

24. Find The distance from a Point $(2, 5, -3)$ To The Plane $\vec{r} \cdot (6\hat{i} - 3\hat{j} + 2\hat{k}) = 5$

$$(\vec{r} - \vec{r}_0) \cdot (\vec{n}) = 0$$

$$6x - 3y + 2z - 5 = 0$$

$$a = 6 \quad b = -3 \quad c = 2 \quad d = -5$$

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$= \frac{|6(2) - 3(5) + 2(-3) - 5|}{\sqrt{6^2 + (-3)^2 + 2^2}}$$

$$= \frac{|12 - 15 - 6 - 5|}{\sqrt{36 + 9 + 4}}$$

$$= \frac{|-14|}{\sqrt{49}} = \frac{14}{7} = 2$$

$D = 2$ units

25 P. T. The function $F(x) = x^2 - 2x - 3$ is strictly increasing in $(2, \infty)$

$$F(x) = x^2 - 2x - 3$$

$$F'(x) = 2x - 2 > 0$$

$F(x)$ is strictly increasing in $(2, \infty)$

26 S. T. The Percentage error

in the n^{th} root of a number is approximately $\frac{1}{n}$ times the percentage error in the number

$$y = x^{1/n}$$

$$\frac{dy}{dx} = \frac{1}{n} x^{(1/n)-1}$$

$$= \frac{1}{n} \cdot \frac{dx}{x} = \frac{y}{nx}$$

$$\frac{dy}{y} = \frac{y}{nx} dx$$

$$\frac{dy}{y} = \frac{1}{n} \left(\frac{dx}{x} \right)$$

$$\left(\frac{dy}{y} \times 100 \right) = \frac{1}{n} \left(\frac{dx}{x} \times 100 \right)$$

Hence Percentage error in the n^{th} root of a number is approximately $\frac{1}{n}$ times the percentage error in the number

27. Evaluate $\int_0^{\infty} x^5 e^{-3x} dx$

$$\int_0^{\infty} x^5 e^{-3x} dx = \frac{5!}{3^6}$$

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}} \quad n=5, a=3$$

28. Assume that a spherical rain drop evaporates at a rate proportional to its surface area differential. i.e., evaporation rate of change of the radius of rain drop

$$\frac{dv}{dt} \propto A \Rightarrow \frac{dv}{dt} = -K A(K, v_0)$$

$$\frac{dv}{dt} = \frac{d}{dt} \left\{ \frac{4}{3} \pi r^3 \right\}$$

$$= \frac{4}{3} \pi \left(3 r^2 \frac{dr}{dt} \right) = 4 \pi r^2 \frac{dr}{dt}$$

$$4 \pi r^2 \frac{dr}{dt} = -K (4 \pi r^2)$$

$$\frac{dr}{dt} = -K$$

$$\frac{dr}{dt} = -K$$

29. A Pair of four dice is rolled one of End The Probability mass Function To get The number of Form

$$S = \{(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)$$

(2,1)

(2,2) (2,3) (2,4) (2,5) (2,6)

(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)

(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)

(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)

(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

Probability mass function

30. Let

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$F(x) = \begin{matrix} 0 & 1 & 2 \\ 2/3 & 1/6 & 1/6 \end{matrix}$$

Two random matrix of the same of some type find AUB and AUB

$$A \cup B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cup \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A \cap B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cap \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

3 Marks

31. Solve The system of linear equations

$$2x + 5y = -2, \quad x + 2y = -3 \text{ using matrix inversion method}$$

$$X = A^{-1}B$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A \quad |A| = \begin{vmatrix} 2 & 5 \\ 1 & 2 \end{vmatrix} = 4 - 5 = -1$$

$$\text{adj } A = \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = -1 \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$= - \begin{bmatrix} 4 + 15 \\ 2 - 6 \end{bmatrix}$$

$$= - \begin{bmatrix} 19 \\ -4 \end{bmatrix}$$

$$= \begin{bmatrix} -19 \\ 4 \end{bmatrix}$$

$$x = -19 \quad y = 4$$

32 State and Prove Triangle inequality.

Statement: For any two complex numbers z_1 and z_2 we have $|z_1 + z_2| \leq |z_1| + |z_2|$

$$|\bar{z}_1 + \bar{z}_2|^2 = (\bar{z}_1 + \bar{z}_2)(z_1 + z_2)$$

$$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$$

$$= z_1 \bar{z}_1 + \bar{z}_1 z_2 + z_1 \bar{z}_2 + z_2 \bar{z}_2$$

$$= |z_1|^2 + (z_1 \bar{z}_2 + \bar{z}_1 z_2) + |z_2|^2$$

$$= |z_1|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2) + |z_2|^2$$

$$\leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2$$

$$= (|z_1| + |z_2|)^2 \leq (|z_1| + |z_2|)^2$$

$$|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$$

Hence Proved

33 Find The value of $\sin^{-1}(\frac{\sin 5\pi}{9} \cos \frac{\pi}{9} + \cos 5\pi \sin \frac{\pi}{9})$

$$\sin^{-1}(\frac{\sin 5\pi}{9} \cos \frac{\pi}{9} + \cos 5\pi \sin \frac{\pi}{9})$$

$$= \sin^{-1}(\sin \frac{5\pi}{9} \cos \frac{\pi}{9} + \cos \frac{5\pi}{9} \sin \frac{\pi}{9})$$

$$= \sin^{-1}(\sin(\frac{5\pi}{9} + \frac{\pi}{9}))$$

$$= \sin^{-1}(\sin(\frac{6\pi}{9}))$$

$$= \sin^{-1}(\sin(\frac{2\pi}{3}))$$

$$= \sin^{-1}(\sin(\pi - \frac{\pi}{3}))$$

$$= \frac{\pi}{3}$$

34. with usual notation in any

Triangle ABC prove by vector method

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Area of Triangle

$$\frac{1}{2} |\vec{BC} \times \vec{CA}| = \frac{1}{2} |\vec{AC} \times \vec{AB}| = \frac{1}{2} |\vec{AB} \times \vec{BC}|$$

$$|\vec{BC} \times \vec{CA}| = |\vec{AC} \times \vec{AB}| = |\vec{AB} \times \vec{BC}|$$

$$ab \sin(\pi - C) = bc \sin(\pi - A) = ca \sin(\pi - B)$$

$$ab(\sin C) = bc(\sin A) = ca(\sin B)$$

$$\div abc$$

$$\frac{ab \sin C}{abc} = \frac{bc \sin A}{abc} = \frac{ca \sin B}{abc}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Hence Proved

35. Find Two Positive numbers whose Sum is 12 and Their Product is maximum

Let x, y be two numbers

$$x + y = 12 \Rightarrow y = 12 - x \quad (1)$$

$$P = xy$$

$$P = x(12 - x)$$

$$P = 12x - x^2$$

$$P' = 0$$

$$12 - 2x = 0$$

$$12 = 2x$$

$$x = 6$$

$$y = 6$$

sub in (1)

$$y = 6$$

The two numbers are 6 & 6

36. If $v(x, y, z) = \log(x^3 + y^3 + z^3)$

find $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z}$

$$v(x, y, z) = \log(x^3 + y^3 + z^3)$$

$$\frac{\partial v}{\partial x} = \frac{1}{x^3 + y^3 + z^3} (3x^2)$$

$$\frac{\partial v}{\partial y} = \frac{1}{x^3 + y^3 + z^3} (3y^2)$$

$$\frac{\partial u}{\partial z} = \frac{1}{x^3 + y^3 + z^3} (3z^2)$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2 + y^2 + z^2)}{x^3 + y^3 + z^3}$$

37. P.T The Point of intersection of The Tangent at t_1 and t_2 on The Parabola $y^2 = 4ax$ is $[at_1 t_2, a(t_1 + t_2)]$

$$y t_1 = x + a t_1^2 \quad \text{--- (1)}$$

$$y t_2 = x + a t_2^2 \quad \text{--- (2)}$$

$$y(t_1 - t_2) = a(t_1^2 - t_2^2)$$

$$y(t_1 - t_2) = a(t_1 - t_2)(t_1 + t_2)$$

$$y = a(t_1 + t_2)$$

Sub in (1)

$$y t_1 = x + a t_1^2$$

$$a(t_1 + t_2) t_1 = x + a t_1^2$$

$$a t_1^2 + a t_1 t_2 = x + a t_1^2$$

$$x = a t_1 t_2$$

The Point of intersection

$$(a t_1 t_2, a(t_1 + t_2))$$

38. Find The Population of a city at any time t given That The rate of increase of Population at that instant and That in a Period of 40 years The Population increased from 3,00,000 To 4,00,000

Let P be denote The Population of a city

$$\frac{dP}{dt} \propto P$$

$$\frac{dP}{dt} = KP$$

$$\frac{dP}{P} = K dt$$

$$\int \frac{dP}{P} = K \int dt$$

$$\log P = Kt + \log C$$

$$\log P - \log C = Kt$$

$$\log \left(\frac{P}{C} \right) = Kt$$

$$\frac{P}{C} = e^{Kt}$$

$$P = C e^{Kt} \quad \text{--- (1)}$$

$$\text{when } t=0 \quad P = 3,00,000$$

$$3,00,000 = C e^{K(0)}$$

$$C = 3,00,000$$

Sub in (1)

$$P = 3,00,000 e^{Kt} \quad \text{--- (2)}$$

$$\text{when } t=40, \quad P = 4,00,000$$

$$4,00,000 = 3,00,000 e^{Kt}$$

$$\frac{4}{3} = e^{40K}$$

$$\log \left(\frac{4}{3} \right) = 40K$$

$$K = \frac{1}{40} \log \left(\frac{4}{3} \right)$$

$$K = \log \left(\frac{4}{3} \right)^{1/40}$$

$$P = 3,00,000 e^{\log \left(\frac{4}{3} \right)^{1/40} \times t}$$

$$= 3,00,000 \left(\frac{4}{3} \right)^{t/40}$$

39. Establish The equivalence property connecting the biconditional with conditional $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$P \leftrightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T
$(P \rightarrow Q) \wedge (Q \rightarrow P)$				
		T	F	F
		F	T	F
		T	T	T

Hence Proved

40 S.T The Polynomial $9x^9 + 2x^5 - 7x^2 + 2$ has at least 8 imaginary roots

$$P(x) = 9x^9 + 2x^5 - 7x^2 + 2$$

No. of sign change = 2

No. of Positive roots = 2

$$P(-x) = -9x^9 - 2x^5 - 7x^2 + 2$$

No. of sign change = 1

No. of Negative roots = 1

No. of Zero = 0

It has at least 8 imaginary roots

5 Marks

41 a) Solve The system of linear Equation by cramer's Rule

$$\begin{aligned} x_1 - x_2 &= 3, & 2x_1 + 3x_2 + 4x_3 &= 17 \\ x_2 + 2x_3 &= 7 \end{aligned}$$

$$\Delta = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{vmatrix}$$

$$= 1(6-4) + 1(4)+0$$

$$\Delta_1 = \begin{vmatrix} 3 & -1 & 0 \\ 17 & 3 & 4 \\ 7 & 1 & 2 \end{vmatrix}$$

$$\begin{aligned} &= 3(6-4) + 1(34-28)+0 \\ &= 3(2) + 1(6) = 6+6=12 \end{aligned}$$

$$\Delta_2 = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 17 & 4 \\ 0 & 7 & 2 \end{vmatrix}$$

$$\begin{aligned} &= 1(34-28) - 3(4)+0 \\ &= 6-12=-6 \end{aligned}$$

$$\Delta_3 = \begin{vmatrix} 1 & -1 & 3 \\ 2 & 3 & 17 \\ 0 & 1 & 7 \end{vmatrix}$$

$$= 1(21-17) + 1(14) + 3(2) \\ = 4 + 14 + 6 = 24$$

$$\frac{\Delta_1}{\Delta} = \frac{12}{6} = 2$$

$$\frac{\Delta_2}{\Delta} = \frac{-6}{6} = -1$$

$$\frac{\Delta_3}{\Delta} = \frac{24}{6} = 4$$

b) Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

$$\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{\frac{x-1}{x+2} + \frac{x+1}{x+2}}{1 - \left(\frac{x-1}{x+2}\right)\left(\frac{x+1}{x+2}\right)}\right) = \frac{\pi}{4}$$

$$\frac{(x+2)(x-1) + (x+1)(x+2)}{(x-2)(x+2) - (x-1)(x+1)} = \frac{\pi}{4}$$

$$\frac{(x-2)(x+2)}{(x-2)(x+2)}$$

$$\frac{(x-2)(x+2) - (x-1)(x+1)}{(x-2)(x+2)}$$

$$\frac{x^2 - x + 2x - 2 + x^2 - 2x + x^2 + 2}{x^2 + 2x - 2x - 4 - x^2 - x + x + 1} = 1$$

$$\frac{2x^2 - 4}{-3} = 1$$

$$2x^2 - 4 = -3$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

42) a) If $Z = x + iy$ and $\arg\left(\frac{Z-i}{Z+i}\right) = \frac{\pi}{4}$

$$= \frac{\pi}{4} \quad \text{S.T. } x^2 + y^2 + 3x - 3y + 2 = 0$$

$$\arg\left(\frac{Z-i}{Z+i}\right) = \frac{\pi}{4}$$

$$\arg\left(\frac{bx + i^2y - i}{cx + i^2y + 2}\right) = \frac{\pi}{4}$$

$$\arg\left(\frac{xc + (y-1)i}{(x+2) + iy}\right) = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{\frac{y-1}{x} - \frac{y}{x+2}}{1 + \left(\frac{y-1}{x} \right) \left(\frac{y}{x+2} \right)} \right) = \frac{\pi}{4}$$

$$\frac{(y-1)(x+2) - xy}{(x)(x+2) + y^2 - y} = \tan \frac{\pi}{4}$$

$$\frac{xy + 2y - x^2 - 2xy}{x^2 + 2x + y^2 - y} = 1$$

$$-x^2 + 2y - 2 = x^2 + y^2 + 2x - y$$

Hence Proved

b) If $u = \sin^{-1} \left(\frac{x+y}{\sqrt{x+y}} \right)$

S.T $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = \frac{1}{2}$

$$u = \sin^{-1} \left(\frac{x+y}{\sqrt{x+y}} \right)$$

u is not homogeneous

Let $F(x, y) = \frac{x+y}{\sqrt{x+y}}$

$$F = \sin u$$

$$F(\lambda x, \lambda y) = \frac{\lambda x + \lambda y}{\sqrt{\lambda x + \lambda y}}$$

$$= \lambda \frac{(\lambda x + \lambda y)}{\sqrt{\lambda} (\sqrt{x+y})}$$

$$= \lambda^{1/2} \frac{(\lambda x + \lambda y)}{\sqrt{x+y}}$$

F is a homogeneous with degree $\frac{1}{2}$

By Euler's formula $x \frac{\partial F}{\partial x} + y \frac{\partial F}{\partial y} = nF$

$$\frac{x \sin u}{\sin x} + y \frac{\sin u}{\sin y} = \frac{1}{2}$$

$$x \cos u \frac{du}{dx} + y \cos u \frac{du}{dy} =$$

$$\frac{1}{2} \sin u$$

$$\div \cos u$$

$$x \frac{du}{dx} + y \frac{du}{dy} = \frac{1}{2} \tan u$$

Hence Proved

$$43 a) \text{ Evaluate } \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$$

$$\text{Let } g(x) = (\sin x)^{\tan x}$$

$$\log g(x) = \log (\sin x)^{\tan x}$$

$$= \tan x \log (\sin x)$$

$$= \log (\sin x)$$

$$\cot x$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \log g(x) = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\log (\sin x)}{\cot x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sin x} \times \cos x}{-\csc^2 x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x / \sin x}{-1 / \sin^2 x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} -\cos x - \sin x$$

$$= 0$$

$$b) \text{ If } \vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}, \vec{b} = 3\vec{i} + 5\vec{j} + 2\vec{k}, \vec{c} = -\vec{i} - 2\vec{j} + 3\vec{k}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a}$$

L.H.S

$$\vec{a} \times \vec{b} =$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & -1 \\ 3 & 5 & 2 \end{vmatrix}$$

$$= \vec{i}(6+5) - \vec{j}(4+3) + \vec{k}(10-9)$$

$$= 11\vec{i} - 7\vec{j} + \vec{k}$$

$$\begin{aligned}
 (\vec{a} \times \vec{b}) \times \vec{c} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -7 \\ 1 & 1 & -2 \end{vmatrix} \\
 &= \hat{i}(-2+2) - \hat{j}(-3+1) + \hat{k}(-2-1) \\
 &= -19\hat{i} - 34\hat{j} - 29\hat{k} \quad \text{--- (1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{R.H.S} \\
 (\vec{a} \cdot \vec{c}) \vec{b} &= (2\hat{i} + 3\hat{j} - \hat{k}) \cdot \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \vec{b} \\
 &= -2 - 6 - 3 = -11(3\hat{i} + 5\hat{j} \\
 &\quad + 2\hat{k}) \\
 (\vec{b} \cdot \vec{c}) \vec{a} &= (3\hat{i} + 5\hat{j} + 2\hat{k}) \cdot \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \vec{a} \\
 &= -3 - 10 + 6
 \end{aligned}$$

$$\begin{aligned}
 &= -3 - 10 + 6 \\
 &= -7(2\hat{i} + 3\hat{j} - \hat{k}) \\
 (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} &= \\
 &= -33\hat{i} - 55\hat{j} - 22\hat{k} + 14\hat{i} \\
 &\quad + 21\hat{j} - 7\hat{k} \\
 &= -19\hat{i} - 34\hat{j} - 29\hat{k} \quad \text{--- (2)}
 \end{aligned}$$

From (1) & (2)
L.H.S = R.H.S

14 a) Solve

The eq $2x^3 + 11x^2 - 9x - 18 = 0$

$$\begin{array}{r|rrrr}
 -1 & 2 & 11 & -9 & -18 \\
 & 0 & -2 & -9 & 18 \\
 \hline
 & 2 & 9 & -18 & 0
 \end{array}$$

$$2x^2 + 9x - 18 = 0$$

$$(x+6)\left(x-\frac{3}{2}\right) = 0$$

$$x = -6, \frac{3}{2}$$

∴ The solution are $-1, -6, \frac{3}{2}$

b) Solve $(1+x^2) \frac{dy}{dx} = 1+y^2$

$$(1+x^2) \frac{dy}{dx} = 1+y^2$$

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$

$$\tan^{-1} y = \tan^{-1} x + c$$

$$\tan^{-1} y - \tan^{-1} x = c$$

$$\tan^{-1} \left(\frac{y-x}{1+xy} \right) = c$$

$$\frac{y-x}{1+xy} = \tan c$$

$$\frac{y-x}{1+xy} = a$$

$$y-x = a(1+xy)$$

45 a) Find The eq of the circle Passing Through The Point $(1,0)$ $(-1,0)$ and $(0,1)$

$$\text{At } (1,0) \quad x^2 + y^2 + 2gx + 2fy + c = 0$$

$$1 + 2g + c = 0$$

$$2g + c = -1 \quad \text{--- (1)}$$

$$\text{At } (-1,0)$$

$$1 - 2g + c = 0$$

$$\text{At } (0,1) \quad -2g + c = -1 \quad \text{--- (2)}$$

$$1 + 2f + c = 0$$

$$2f + c = -1 \quad \text{--- (3)}$$

① & ②

$$\begin{aligned} 2g + c &= -1 \\ -2g + c &= -1 \\ \hline 2c &= -2 \end{aligned}$$

$$c = -1$$

$$c = -1$$

$c = -1$ Sub in ①

$$2g - 1 = -1$$

$$2g = -1 + 1$$

$$2g = 0$$

$$g = 0$$

$$f = 0$$

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$x^2 + y^2 - 1 = 0$$

b) The

mean and $x^2 + y^2 = 1$

a binomial variate x variance of x is 2 and 1.5 respectively find i) $P(x=0)$ and ii) $P(x=1)$

$$P(x=1) \quad P(x \geq 1)$$

$$nP = 2$$

$$nPv = 1.5$$

$$\frac{nPv}{nP} = \frac{3/2}{2}$$

$$v = \frac{3}{4}$$

$$P = 1 - v$$

$$= 1 - \frac{3}{4}$$

$$nP = 2$$

$$n\left(\frac{1}{4}\right) = 2$$

$$n = 8$$

$$P(x=x) = \binom{n}{x} P^x v^{n-x}$$

$$i) P(x=0) = \binom{8}{0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^8$$

$$= \left(\frac{3}{4}\right)^8$$

$$ii) P(x=1) = \binom{8}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^7$$

$$= 8 \times \frac{1}{4} \left(\frac{3}{4}\right)^7$$

$$= 2 \left(\frac{3}{4}\right)^7$$

$$iii)$$

$$P(x \geq 1) = 1 - P(x=0)$$

$$= 1 - \left(\frac{3}{4}\right)^8$$

$$= 1 - \left(\frac{3}{4}\right)^8$$

4b) Find the foot of the perpendicular from point (5, 4, 2) to the line $\frac{x-1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$

The perpendicular

$$\frac{2x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1} = \lambda$$

$$F = (2\lambda - 1, 3\lambda + 3, -\lambda + 1)$$

$$\vec{AF} = F - A = (2\lambda - 1) - 5, (3\lambda + 3) - 4,$$

$$\vec{AF} = (2\lambda - 6, 3\lambda - 1, -\lambda - 1)$$

$$= 2(2\lambda - 6) + 3(3\lambda - 1) + 1(-\lambda - 1)$$

$$4\lambda - 12 + 9\lambda - 3 + (-\lambda - 1) = 0$$

$$14\lambda - 16 = 0$$

$$\lambda = 1$$

$$F = (2(1) - 1, 3(1) + 3, -1 + 1) = (1, 6, 0)$$

$$A = (5, 4, 2) \text{ and } (1, 6, 0)$$

$$d = \sqrt{(5-1)^2 + (4-6)^2 + (2-0)^2} = \sqrt{16+4+4} = \sqrt{24} = 2\sqrt{6}$$

conversion cal

$$\frac{x-5}{2} = \frac{y-4}{-1} = \frac{z-2}{1}$$

b) Find the area of the line bounded by the region $5x - 2y = 15$, $x + y + 4 = 0$ and x axis using integration

region $5x - 2y = 15$, $x + y + 4 = 0$ and x axis using integration

$$A = \left| \int_{-4}^1 y dx \right| + \left| \int_1^3 y dx \right|$$

$$= \left| \int_{-4}^1 (-4-x) dx \right| + \left| \int_1^3 \left(\frac{5x-15}{2} \right) dx \right|$$

$$= \left| \left[-4x - \frac{x^2}{2} \right]_{-4}^1 \right| + \left| \left[\frac{5x^2}{2} - 15x \right]_1^3 \right|$$

$$= \left| \left(\frac{-9}{2} \right) - 8 \right| + \left| \frac{-45}{4} - \frac{25}{4} \right|$$

$$= \frac{25}{2} + 5 = 3\frac{5}{2}$$

$$A = \int_{-5}^0 (x_R - x_L) dy = \int_{-5}^0 \left[\frac{15+2}{5} - (-4-\frac{y}{2}) \right] dy$$

$$= \int_{-5}^0 \left[7 + \frac{y}{5} \right] dy = \left[7y + \frac{y^2}{10} \right]_{-5}^0$$

$$= 0 - \left[-35 + \frac{35}{2} \right] = \frac{35}{2}$$

47
a) A Particle moves along a line according to the law

$$s(t) = 2t^3 - 9t^2 + 12t - 4$$

At what time the Particle

i) changes direction?

ii) Find the total distance travelled

iii) Find the Particle in the first

by the

4 second

iv) Find the Particle's acceleration

each time the velocity is zero

$$s(t) = 2t^3 - 9t^2 - 4 + 12t$$

$$i) v(t) = 6t^2 - 18t + 12$$

$$= 6(t^2 - 3t + 2)$$

$$v(t) = 6(t-2)(t-1)$$

$$6(t-1)(t-2) = 0$$

$$(t-1)(t-2)=0$$

$$t = 1, 2$$

$$s(0) = -4$$

$$s(1) = 2(1)^3 - 9(1)^2 + 12(1) = 2 - 9 + 12 - 4 = 1$$

$$s(2) = 2(2)^3 - 9(2)^2 + 12(2) = 16 - 36 + 24 - 4 = 0$$

$$s(4) = 2(4)^3 - 9(4)^2 + 12(4) = 128 - 144 + 48 - 4 = 28$$

$$s(0) - s(1) + s(1) - s(2) + s(2) - s(4) = -4 - 1 + 1 - 0 + 4 - 28 = -28$$

$$= -5 + 1 + 28 = 34$$

$$A = 12t - 18$$

$$At = 1, A = 12(1) - 18 = -6$$

$$At = 2, A = 12(2) - 18 = 6 \text{ m/sec}^2$$

b)

i) Closure Property

$$x = \begin{pmatrix} x & x \\ x & x \end{pmatrix}, y = \begin{pmatrix} y & y \\ y & y \end{pmatrix}$$

$$x * y = \begin{pmatrix} x & x \\ x & x \end{pmatrix} * \begin{pmatrix} y & y \\ y & y \end{pmatrix} = \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix}$$

closure is true

Commutative

$$x * y = \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix}$$

$$y * x = \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix}$$

$$x * y = y * x$$

Commutative Property is

True

Associative

$$x * (y * z) = \begin{pmatrix} 4xyz & 4xyz \\ 4xyz & 4xyz \end{pmatrix}$$

$$(x * y) * z = \begin{pmatrix} 4xyz & 4xyz \\ 4xyz & 4xyz \end{pmatrix}$$

Associative Property is True

Identity Property

$$X * E = X$$

$$\begin{pmatrix} x & x \\ x & x \end{pmatrix} * \begin{pmatrix} e & e \\ e & e \end{pmatrix} = \begin{pmatrix} x & x \\ x & x \end{pmatrix}$$

$$\begin{pmatrix} 2x & e \\ 2x & e \end{pmatrix} = \begin{pmatrix} x & x \\ x & x \end{pmatrix}$$

$$2xe = x$$

$$2e = 1$$

$$e = 1/2$$

$$E = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$\therefore$ Identity Property is True

Inverse

$$X * X^{-1} = E$$

$$\begin{pmatrix} x & x \\ x & x \end{pmatrix} * \begin{pmatrix} x^{-1} & x^{-1} \\ x^{-1} & x^{-1} \end{pmatrix} = \begin{pmatrix} e & e \\ e & e \end{pmatrix}$$

$$\begin{pmatrix} 2xx^{-1} & 2xx^{-1} \\ 2xx^{-1} & 2xx^{-1} \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

$$2xx^{-1} = 1/2$$

$$2x^{-1} = 1/4x$$

$$x^{-1} = 1/4x$$

$$\begin{pmatrix} 1/4x & 1/4x \\ 1/4x & 1/4x \end{pmatrix}$$

Inverse Property is True

2 Marks

$$2) \text{ If } |Z| = 2 \text{ s.t. } 3 \leq |Z + 3 + 4i| \Rightarrow$$

$$|Z + 3 + 4i| \leq |Z| + |3 + 4i|$$

$$\leq 2 + \sqrt{3^2 + 4^2}$$

$$= 2 + \sqrt{9 + 16}$$

$$= 2 + \sqrt{25}$$

$$= 2 + 5$$

$$= 7 \text{ --- ①}$$

$$|Z + 3 + 4i| \geq |Z| - |3 + 4i|$$

$$\geq |2| - \sqrt{3^2 + 4^2}$$

$$\geq |2| - \sqrt{9 + 16}$$

$$\geq |2| - \sqrt{25}$$

$$\geq |2| - 5$$

$$\geq |-3|$$

$$F \text{ from ① } \geq 3 \text{ --- ②}$$

$$3 \leq |Z + 3 + 4i| \Rightarrow$$

Hence Proved

22 If P and q are the roots of

$$\text{The eq } lx^2 + mx + n = 0 \text{ s.t.}$$

$$\sqrt{\frac{p}{a}} + \sqrt{\frac{q}{a}} + \sqrt{\frac{n}{l}} = 0$$

$$\text{Sum of roots} = -\frac{b}{a}$$

$$P+q = -\frac{n}{l}$$

$$\text{Product of roots} = \frac{c}{a}$$

$$Pq = \frac{n}{l}$$

$$\sqrt{\frac{p}{a}} + \sqrt{\frac{q}{a}} = \sqrt{\frac{P}{a}} + \sqrt{\frac{q}{a}} = \frac{P+q}{\sqrt{Pa}}$$

$$= \frac{P+q}{\sqrt{Pa}}$$

$$\frac{P+q}{\sqrt{Pa}} = -\frac{n}{l}$$

$$\sqrt{\frac{P}{a}} + \sqrt{\frac{q}{a}} = -\sqrt{\frac{n}{l}}$$

$$\left(\sqrt{\frac{P}{a}} + \sqrt{\frac{q}{a}}\right)^2 = \left(-\sqrt{\frac{n}{l}}\right)^2$$

25.

23

If $y = 4x + c$ is a tangent

To the circle $x^2 + y^2 = 9$ Find

$$y = mx + c$$

$$x^2 + y^2 = a^2$$

$$y = 4x + c$$

$$x^2 + y^2 = 9$$

$$m = 4$$

$$a^2 = 9$$

24

If the radius of the sphere with radius 10 cm, has to decrease by 0.1 cm approximately how much will its volume decrease

$$V = \frac{4}{3} \pi r^3$$

$$dV = \frac{4}{3} \pi (3r^2) dr$$

$$= 4 \pi (10)^2 (-0.1)$$

$$= -40 \pi \text{ cm}^3$$

Evaluate

20

$$\int_a^b \frac{1}{a^2+x^2} dx \quad a > 0, b \in \mathbb{R}$$

$$\int_a^b \frac{1}{a^2+x^2} dx = \left[\frac{1}{a} \tan^{-1} \frac{x}{a} \right]_a^b$$

$$\begin{aligned} C^2 &= a^2 (1+m^2) \\ &= 9 (1+16) \\ &= 9 \times 17 \\ C &= \sqrt{9 \times 17} \\ C &= \pm 3 \sqrt{17} \end{aligned}$$

$$= \frac{1}{a} \left[\tan^{-1}(\infty) - \tan^{-1}\left(\frac{b}{a}\right) \right]$$

$$= \frac{1}{a} \left[\frac{\pi}{2} - \tan^{-1}\frac{b}{a} \right]$$

26. Find the vector eq. of a plane which is at a distance of 7 units from the origin having 3, -4, 5 as direction ratios of a normal to it

$$\hat{d} = \frac{3\hat{i} - 4\hat{j} + 5\hat{k}}{\sqrt{3^2 + (-4)^2 + 5^2}} = \frac{3\hat{i} - 4\hat{j} + 5\hat{k}}{\sqrt{50}}$$

$$\hat{r} \cdot \hat{d} = p \quad \Rightarrow \quad \frac{\hat{r} \cdot (3\hat{i} - 4\hat{j} + 5\hat{k})}{\sqrt{50}} = 7$$

$$\hat{r} \cdot \left(\frac{3\hat{i} - 4\hat{j} + 5\hat{k}}{\sqrt{50}} \right) = 7$$

27 Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

be any two basis matrices of the same type Find $A \cup B$ and $A \cap B$

$$A \cup B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cup \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A \cap B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \cap \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

28. P.T $\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ is orthogonal

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \quad A^T = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$AA^T = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & \cos \alpha \sin \alpha - \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha - \sin \alpha \cos \alpha & \sin^2 \alpha + \cos^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$A^T A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\sin \alpha \cos \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha + \sin \alpha \cos \alpha & \cos^2 \alpha + \sin^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$A A^T = A^T A = I_2$$

A is orthogonal

Here Proved

29. Find the Eq of Tangent

To the curve $y = x^2 + 3x - 2$

at the Point $(1, 2)$

$$y = x^2 + 3x - 2$$

D.W.8

$\frac{dy}{dx}$

$$= 2x + 3$$

At Point $(1, 2)$

$$2(1) + 3 = 5$$

$$m = 5$$

Eq of Tangent $y - y_1 = m(x - x_1)$

$$y - 2 = 5(x - 1)$$

$$y - 2 = 5x - 5$$

$$5x - y - 5 + 2 = 0$$

$$5x - y - 3 = 0$$

30 Express $e^{\cos \alpha + i \sin \alpha}$ in $a + ib$ form

$$e^{\cos \alpha + i \sin \alpha} = e^{\cos \alpha} \cdot e^{i \sin \alpha}$$

$$(e^{i x} = \cos x + i \sin x, \text{ where } x = \sin \alpha)$$

$$e^{i \sin \alpha} = \cos(\sin \alpha) + i \sin(\sin \alpha)$$

$$= e^{\cos \alpha} \cos(\sin \alpha) + i e^{\cos \alpha} \sin(\sin \alpha)$$

3 marks

31 Find the Eq of the Parabola with vertex $(-1, -2)$ and Parallel to y-axis and Passing Through $(3, 6)$

To y-axis and Passing Through $(3, 6)$

$$A^T A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\sin \alpha \cos \alpha + \sin \alpha \cos \alpha \\ -\sin \alpha \cos \alpha + \sin \alpha \cos \alpha & \cos^2 \alpha + \sin^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$A A^T = A^T A = I_2$$

A is orthogonal

Hence Proved

29. Find the Eq. of Tangent

To the curve $y = x^2 + 3x - 2$ at the Point $(1, 2)$

$$y = x^2 + 3x - 2$$

D. W. 8

$$\frac{dy}{dx}$$

$$= 2x + 3$$

At Point $(1, 2)$

$$2(1) + 3 = 5$$

$$m = 5$$

Eq. of Tangent

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 5(x - 1)$$

$$y - 2 = 5x - 5$$

$$5x - y - 5 + 2 = 0$$

$$5x - y - 3 = 0$$

30 Express $e^{\cos \alpha + i \sin \alpha}$

$$e^{\cos \alpha + i \sin \alpha} = e^{\cos \alpha} \cdot e^{i \sin \alpha} \quad \text{in } a + ib \text{ form}$$

$$(e^{ix} = \cos x + i \sin x, \text{ where } x = \sin \alpha)$$

$$e^{i \sin \alpha} = \cos(\sin \alpha) + i \sin(\sin \alpha)$$

$$= e^{\cos \alpha} \cos(\sin \alpha) + i e^{\cos \alpha} \sin(\sin \alpha)$$

3 Marks

31 Find the Eq. of the Parabola with vertex $(-1, -2)$ and Parallel to y-axis and Passing Through $(3, 6)$

To y-axis and Passing Through $(3, 6)$

$$(x-h)^2 = 4a(y-k)$$

$$(3+1)^2 = 4a(6+2)$$

$$16 = 4a(8)$$

$$\div 8$$

$$2 = 4a$$

$$a = \frac{2}{4} = \frac{1}{2}$$

$$(x+1)^2 = 2(y+2)$$

$$x^2 + 2x + 1 = 2y + 4$$

$$x^2 + 2x - 2y - 3 = 0$$

32

The maximum and minimum distance of the Earth from the Sun respectively are 152×10^6 km and 94.5×10^6 km. The Sun is at one focus of the elliptical orbit. Find the distance from the Sun to the other focus.

$$a + c = 152 \times 10^6$$

$$a - c = 94.5 \times 10^6$$

33.

$$= 575 \times 10^6$$

Distance from the Sun

$$= 575 \times 10^6 \text{ km}$$

For what value of x , the inequality $\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$ holds?

$$\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$$

$$\cos \frac{\pi}{2} < 3x-1 < \cos \pi$$

$$0 < 3x-1 < -1$$

$$0+1 < 3x < -1+1$$

$$\frac{1}{3} < x < 0$$

The inequality is true only when $0 < x < \frac{1}{3}$

34.

Find the angle made by straight line $\frac{x+3}{2} = \frac{y-1}{2} = -z$ with coordinate axes

$$\frac{x+3}{2} = \frac{y-1}{2} = -z$$

$$(x-h)^2 = 4a(y-k)$$

$$(3+1)^2 = 4a(6+2)$$

$$16 = 4a(8)$$

$$\div 8$$

$$2 = 4a$$

$$a = \frac{2}{4} = \frac{1}{2}$$

$$(x+1)^2 = 2(y+2)$$

$$x^2 + 2x + 1 = 2y + 4$$

$$x^2 + 2x - 2y - 3 = 0$$

32

The maximum and minimum distance of The Earth From The Sun respectively are 152×10^6 km and 94.5×10^6 km. The Sun is at one focus of the elliptical orbit. Find The distance from The Sun to the other focus

$$a + c = 152 \times 10^6$$

$$a - c = 94.5 \times 10^6$$

$$\hline 2c = 57.5 \times 10^6$$

$$= 57.5 \times 10^6$$

Distance from The Sun = 57.5×10^6 km

33.

For what value of x , the inequality $\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$ holds,

$$\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$$

$$\cos \frac{\pi}{2} < 3x-1 < \cos \pi$$

$$0 < 3x-1 < -1$$

$$0+1 < 3x < -1+1$$

$$\frac{1}{3} < x < 0$$

The inequality is true only when $0 < x < \frac{1}{3}$

34.

Find The angle made by straight line $\frac{x+3}{2} = \frac{y-1}{2} = -2$ with coordinate axis

$$\frac{x+3}{2} = \frac{y-1}{2} = -2$$

$$\vec{b} = 2\hat{i} + 2\hat{j} - \hat{k}$$

$$|\vec{b}| = \sqrt{2^2 + 2^2 + (-1)^2}$$

$$= \sqrt{4+4+1} = \sqrt{9} = 3$$

$$= \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$$

$$\cos \alpha = \frac{2}{3}$$

$$\alpha = \cos^{-1}\left(\frac{2}{3}\right) \quad \left| \quad \cos \beta = \frac{2}{3} \right.$$

$$\cos \gamma = -\frac{1}{3} \quad \left| \quad \beta = \cos^{-1}\left(\frac{2}{3}\right) \right.$$

$$\gamma = \cos^{-1}\left(-\frac{1}{3}\right)$$

35 Use the linear approximation
To find an approximate value
 $(1.23)^{2/3}$

$$f(x) = x^{2/3}$$

$$f'(x) = \frac{2}{3} x^{-1/3}$$

$$= \frac{2}{3} x^{-1/3}$$

$$= \frac{2}{3 \times 13}$$

$$f(x_0) = (1.25)^{2/3}$$

$$= (5/4)^{2/3}$$

$$= 5^2 = 25$$

$$f'(x_0) = \frac{2}{3}$$

$$= \frac{2}{3(5)^{1/3} 3(5)} = \frac{2}{15}$$

$$= \frac{2}{15}$$

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$= 25 + \frac{2}{15} (1.23 - 1.25)$$

$$= 25 + \frac{2}{15} - 2$$

$$= 25 - 0.27 = 24.73$$

36. Solve

$$x \cos y \, dy = e^x (x \log x + 1) dx$$

$$x \cos y \, dy = e^x (x \log x + 1) dx$$

$$= e^x \left(\log x + \frac{1}{x} \right) dx$$

$$\int \cos y \, dy = \int e^x \left(\log x + \frac{1}{x} \right) dx$$

$$\sin y = e^x \log x + c$$

37. If $F(x) = \begin{bmatrix} \cos x & 0 & \sin x \\ 0 & 1 & 0 \\ -\sin x & 0 & \cos x \end{bmatrix}$ S.T

$$[F(x)]^{-1} = F(-x)$$

$$F(\alpha)^{-1} = \frac{1}{|F(\alpha)|} \text{adj}[F(\alpha)]$$

$$|F(\alpha)| = \cos \alpha (\cos \alpha) - 0 + \sin \alpha (\sin \alpha)$$

$$= \sin^2 \alpha + \cos^2 \alpha = 1$$

cofactor of $F(\alpha)$

$$\text{adj}[F(\alpha)] = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

$$[F(\alpha)]^{-1} = \frac{1}{1} \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

$$F(-\alpha) = \begin{bmatrix} \cos(-\alpha) & 0 & \sin(-\alpha) \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix} \quad \text{--- (2)}$$

From (1) & (2)

$$[F(\alpha)]^{-1} = F(-\alpha)$$

Hence proved

38. T P $\rightarrow$ q and q $\rightarrow$ p are not equivalent

P	q	P $\rightarrow$ q	q $\rightarrow$ p
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

P $\rightarrow$ q and q $\rightarrow$ p are not equivalent

39. If $Z = (2+3i)(1-i)$ Then find Z^{-1}

$$Z = (2+3i)(1-i)$$

$$= 2 - 2i + 3i + 3$$

$$= 5 + i$$

$$Z^{-1} = \frac{1}{Z} = \frac{1}{5+i} \times \frac{5-i}{5-i}$$

$$= \frac{5}{26} - \frac{i}{26}$$

$$A = b + c - a$$

$$B = c + a - b$$

$$C = a + b - c$$

$$A = -a - a = -2a$$

$$B = -b - b = -2b$$

$$C = -c - c = -2c$$

$$\Delta = b^2 - 4ac$$

$$= (-2b)^2 - 4(-2a)(-2c)$$

$$= 4b^2 - 16ac$$

$$\text{Sqrt } b = -(b+c)$$

$$\Delta = 4(-(b+c))^2 - 16ac$$

$$= 4a^2 + 8ac + 4c^2 - 16ac$$

$$= 4a^2 - 8ac + 4c^2$$

$$= 4(a^2 - 2ac + c^2)$$

$$= 4(a-c)^2$$

$$\Delta = [2(a-c)]^2$$

a, b, c are rational numbers $\Rightarrow (a-c)$ is rational

5 Marks

4) a) Solve the eq $z^3 + 8i = 0$
where $z \in \mathbb{C}$

$$z^3 + 8i = 0$$

$$z^3 - (-2i)^3 = 0$$

$$(z - 2i)(z^2 + 2zi - 4) = 0$$

$$z - 2i = 0$$

$$z = 2i$$

$$z^2 + 2zi - 4 = 0$$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2}$$

$$= \frac{-2i \pm \sqrt{2}}{2}$$

$$= \frac{-2i \pm 2\sqrt{2}}{2}$$

$$= \frac{-i \pm \sqrt{2}}{1}$$

$$z = \sqrt{3} - i, -\sqrt{3} - i$$

The roots are $2i, \sqrt{3} - i, -\sqrt{3} - i$

b) solve $(1+x+xy^2) \frac{dx}{dy} + (y-y^3)x = 0$

$$(1+x+xy^2) \frac{dx}{dy} + (y+y^3) = 0$$

$$(1+x+xy^2) \frac{dx}{dy} + (y+y^3) = 0$$

$$\frac{1+x+xy^2}{y+y^3} + \frac{dy}{dx} = 0$$

$$\frac{dx}{dy} + \frac{1+x(1+y^2)}{y+y^3} = 0$$

$$\frac{dx}{dy} + \frac{1}{y+y^3} + \frac{x(1+y^2)}{y+y^3} = 0$$

$$\frac{dx}{dy} + \frac{x(1+y^2)}{y(1+y^2)} = -\frac{1}{y(1+y^2)}$$

$$\frac{dx}{dy} + \frac{x}{y} = -\frac{1}{y(1+y^2)}$$

This is a linear differential eq

$$P = \frac{1}{y}, Q = -\frac{1}{y(1+y^2)}$$

$$e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x}$$

$$x e^{\int P dx} = \int Q e^{\int P dx} dy + C$$

$$x y = \int \frac{-1}{y(1+y^2)} y dy + C$$

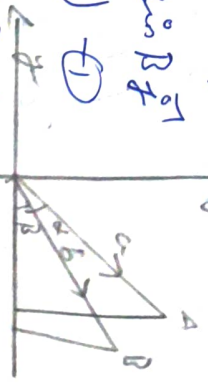
$$x y = - \int \frac{1}{1+y^2} dy + C$$

$$x y = -\tan^{-1} y + C$$

$$x y + \tan^{-1} y = C$$

a) using vector method

$$\begin{aligned} \vec{a} &= \cos \alpha \vec{i} + \sin \alpha \vec{j} \\ \vec{b} &= \cos \beta \vec{i} + \sin \beta \vec{j} \\ \vec{a} \cdot \vec{b} &= \cos(\alpha - \beta) \quad \text{--- (1)} \\ \vec{a} \cdot \vec{b} &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \end{aligned}$$



From (1) & (2)

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Hence Proved

b)

$$i) \int_{200}^{600} K dx = 1$$

$$K [x]_{200}^{600} = 1$$

$$K(600-200) = 1$$

$$400K = 1$$

$$K = \frac{1}{400}$$

$$ii) \text{ For } x < 200$$

$$F(x) = 0$$

$$\text{For } 200 \leq x \leq 600$$

$$F(x) = \int_{200}^x \frac{1}{400} dt = \frac{1}{400} [t]_{200}^x$$

$$= \frac{x-200}{400}$$

$$\text{For } x > 600$$

$$F(x) = 1$$

$$F(x) = \int_{200}^x \frac{1}{400} dt$$

$$= \frac{1}{400} [t]_{200}^x = \frac{x-200}{400}$$

$$iii) P(300 \leq x \leq 500) = F(500) - F(300)$$

$$F(x) = \frac{x-200}{400}$$

$$F(500) = \frac{500-200}{400} = \frac{300}{400} = \frac{3}{4}$$

$$F(300) = \frac{300-200}{400} = \frac{100}{400} = \frac{1}{4}$$

$$F(300 \leq x \leq 500) = \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

143 a) Identify The type of conic

and find centre, foci and

$$\text{Vertices of } 18x^2 + 12y^2 - 144x + 48y + 120 = 0$$

$$h = -\frac{-144}{2 \times 18} = 4$$

$$k = -\frac{48}{2 \times 12} = -2$$

$$h = 4, k = -2$$

$$h = 4, k = -2$$

$$18(4)^2 + 12(2)^2 = 120$$

$$288 + 48 - 120 = 216$$

$$\frac{18(4-4)^2}{216^{1/2}} + \frac{12(4+2)^2}{216^{1/2}} = \frac{216}{216}$$

$$a^2 = 18 \quad | \quad b^2 = 12$$

$$a = 3\sqrt{2} \quad | \quad b = 2\sqrt{3}$$

Centre $(h, k) \quad (4, -2)$

foci $(h, k \pm ae) \quad (4, -2 \pm \sqrt{2})$

vertices $(h, k \pm a) \quad (4, -2 \pm 3\sqrt{2})$

$(4, -2 - 3\sqrt{2})$

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$

and $0 < x, y, z < 1$ So T

$$x^2 + y^2 + z^2 + 2xyz = 1$$

$$\cos^{-1}x = \alpha \quad | \quad \cos^{-1}y = \beta$$

$$x = \cos \alpha \quad | \quad y = \cos \beta$$

$$\alpha + \beta + \cos^{-1}z = \pi$$

$$\alpha + \beta = \pi - \cos^{-1}z$$

$$\cos(\alpha + \beta) = \cos(\pi - \cos^{-1}z)$$

$\cos \alpha \cos \beta - \sin \alpha \sin \beta = \cos(\alpha + \beta)$

$$xy - \sqrt{1-x^2} \sqrt{1-y^2} = \cos(\alpha + \beta)$$

$$xy + z = \sqrt{1-x^2} \sqrt{1-y^2}$$

So on both sides

$$(xy + z)^2 = (1-x^2)(1-y^2)$$

$$x^2y^2 + 2xyz + z^2 = 1 - x^2 - y^2 + x^2y^2$$

$$x^2 + y^2 + 2xyz + z^2 = 1$$

H.P

A boy is walking along the path

The point $(-6, 8)$ through

he wants to meet his friend

P(7, 6) will he meet his friend

$$y = ax^2 + bx + c$$

At $(6, 8)$

$$8 = a(-6)^2 + b(-6) + c$$

$$8 = 36a - 6b + c \quad \text{--- (1)}$$

$$At (-2, -12)$$

$$-12 = a(-2)^2 + b(2) + c$$

$$-12 = 4a - 2b + c \quad (2)$$

$$At (3, 8)$$

$$8 = a(3)^2 + b(3) + c$$

$$8 = 9a + 3b + c \quad (3)$$

$$\begin{bmatrix} 36 & -6 & 1 \\ 4 & -2 & 1 \\ 9 & 3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 8 \\ -12 \\ 8 \end{bmatrix}$$

$$AX = B$$

$$(A, B) \sim \begin{bmatrix} 36 & -6 & 1 & 8 \\ 4 & -2 & 1 & -12 \\ 9 & 3 & 1 & 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 9R_1$$

$$R_3 \rightarrow 4R_3 - 9R_1$$

$$\sim \begin{bmatrix} 4 & -2 & 1 & -12 \\ 0 & 12 & -8 & 116 \\ 0 & 30 & -5 & 140 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2/4 \\ R_3 \rightarrow R_3/5 \end{array} \sim$$

$$\begin{bmatrix} 4 & -2 & 1 & -12 \\ 0 & 3 & -2 & 29 \\ 0 & 0 & -3 & -20 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -2 & 1 \\ 0 & 3 & -2 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -12 \\ 29 \\ -30 \end{bmatrix}$$

$$0 + 0 + 3c = -30$$

$$c = \frac{-30}{3}$$

$$3b - 2c = 29 \quad c = -10$$

$$3b + 20 = 29 \quad 4a - 2b + c =$$

$$3b = 29 - 20 \quad 4a - 2(3) - 10 = -1$$

$$3b = 9 \quad 4a = -12 + 16$$

$$b = 3 \quad a = \frac{4}{4}$$

$$y = ax^2 + bxc + c$$

$$y = 1x^2 + 3xc + 10$$

$$y = x^2 + 3x - 10$$

$$At (7, 60)$$

$$60 = 7^2 + 3(7) - 10$$

$$60 = 49 + 21 - 10$$

$$80 = 60$$

Yes he will meet his friend

b) P.T ellipse $x^2 + 4y^2 = 8$ and the hyperbola $x^2 - 2y^2 = 4$ intersect orthogonally

Let the Point of intersection (a, b)

$$x^2 + 4y^2 = 8$$

$$2x + 8y \frac{dy}{dx} = 0$$

$$\div 2$$

$$x + 4y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-x}{4y}$$

$$m = \frac{-a}{4b}$$

$$m_1 m_2 = \frac{-a}{4b} \times \frac{a}{2b}$$

$$= -\frac{a^2}{8b^2}$$

①

$$a^2 + 4b^2 - 8 \text{ and } a^2 - 2b^2 - 4$$

$$\begin{pmatrix} 1 & 4 & 1 \\ -2 & 8 & 1 \\ -4 & -4 & 1 \end{pmatrix}$$

$$\frac{a^2}{-16-16} = \frac{b^2}{-8+4} = \frac{1}{-2-4}$$

$$\frac{a^2}{-32} = \frac{-b^2}{-4}$$

$$\frac{a^2}{b^2} = \frac{32}{4} = 8$$

$$m_1 m_2 = \frac{-8}{8} = -1$$

Hence the curves cut orthogonally

45. a) Find the Parametric form of vector eq and cartesian

eq of the Plane containing the line $\vec{r} = (\vec{i} - \vec{j} + 3\vec{k}) + t$

$2\vec{i} - \vec{j} + 4\vec{k}$ and perpendicular to the Plane $\vec{r} = \vec{i} + 2\vec{j} + \vec{k}$ is 8

$$\vec{a} = \vec{i} - \vec{j} + 3\vec{k}, \vec{b} = 2\vec{i} - \vec{j} + 4\vec{k}$$

$$\begin{pmatrix} 2x-1 & y+1 & z-3 \\ 2 & -1 & 4 \end{pmatrix} = 0$$

$$(x-1)(-9) - (y+1)(-2) + (z-3)(5) = 0$$

$$-9x + 9 + 2y + 2 + 5z - 15 = 0$$

$$-9x + 2y + 5z - 4 = 0$$

x by -

$$9x - 2y - 5z + 4 = 0$$

$$9x - 2y - 5z = -4$$

Parametric Equation

$$\vec{r} = \vec{a} + s\vec{b} + t\vec{c}$$

$$= 1 - s + 3t + 5(2s - 7t + 4t) + t(\vec{a} + 2\vec{b} + \vec{c})$$

b) Solve The Eq

$$-5x + 6 = 0 \text{ if it is known that } \frac{1}{3}$$

is a solution

$$\frac{1}{3} \quad 6 \quad -5 \quad -38 \quad -5 \quad 6$$

$$\begin{array}{r|rrrrrr} 3 & 6 & -3 & -39 & -18 & 10 \\ & 0 & 18 & 45 & 18 & \\ \hline & 6 & 15 & 6 & 0 & \\ \hline 6x^2 + 15x + 6 = 0 \end{array}$$

$$(x+2)(x+\frac{1}{2}) = 0$$

$$x = -2, -\frac{1}{2}$$

The solution are $\frac{1}{3}, 3, -2, -\frac{1}{2}$

4b a) P, T P $\rightarrow$ (True $\vee$ False) $\equiv$ P $\vee$ (True $\vee$ False) using Truth Table

P	Q	R	$\neg Q$	$\neg R$	$\neg Q \vee \neg R$	$P \rightarrow (\neg Q \vee \neg R)$
T	T	T	F	F	F	F
T	T	F	F	T	T	T
T	F	T	T	F	T	T
T	F	F	T	T	T	T
F	T	T	F	F	F	T
F	T	F	F	T	T	T
F	F	T	T	F	T	T
F	F	F	T	T	T	T

$\rightarrow$ P $\vee$ $\neg Q \vee \neg R$

T F T T T T T

Hence Proved

b) Suppose a person deposits
 £1000 in a bank account at
 The rate of 5% per annum
 compounded continuously. How
 much money will be in his/her
 account 18 months later?

Let P be the Principal at
 Time t

$$r = 5\%$$

$$\frac{dP}{dt} = P \left(\frac{5}{100} \right)$$

$$\frac{dP}{dt} = 0.05P$$

$$\int \frac{dP}{P} = 0.05 \int dt$$

$$\int \frac{dP}{P} = 0.05 t \log e$$

$$\log P = \log C = 0.05 t$$

$$\log \left(\frac{P}{C} \right) = 0.05 t$$

$$\frac{P}{C} = e^{0.05 t}$$

$$P = C e^{0.05 t} \quad (1)$$

$$\text{when } t=0, P=10,000$$

$$10,000 = C e^0$$

a) Find the maximum value of
 $\frac{\log x}{x}$

$$\left(\frac{u}{v} \right)' = \frac{u'v - uv'}{v^2}$$

$$F'(x) = \left(\frac{1}{x} \right) (x) - \log(x) (1)$$

$$x^2$$

$$F'(x) = 0$$

$$1 - \log x = 0$$

$$\log x = 1$$

$$x = e$$

$$F''(x) = \left(-\frac{1}{x} \right) (x^2) - (1 - \log x)$$

$$\text{Sub } x = e$$

$$F''(e) = \frac{-3 + 2 \log e}{e^3} = \frac{-3 + 2(1)}{e^3}$$

$$= \frac{-1}{e^3} < 0$$

Maximum value

$$F(e) = \frac{\log e - 1}{e}$$

b) Find the area of the region common to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ and the straight line}$$

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{x}{a} + \frac{y}{b} = 1 \rightarrow \frac{y}{b} = 1 - \frac{x}{a}$$

$$y = b \left(1 - \frac{x}{a} \right)$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} = \frac{a^2 - x^2}{a^2}$$

$$y_1 = \frac{b}{a} \sqrt{a^2 - x^2}$$

$$y_2 = \frac{b}{a} (a - x)$$

$$A = \int_0^a (y_1 - y_2) dx$$

$$A = \int_0^a \left[\frac{b}{a} \sqrt{a^2 - x^2} - \frac{b}{a} (a - x) \right] dx$$

$$A = \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx - \int_0^a \frac{b}{a} (a - x) dx$$

$$1) \int \sqrt{a^2 - x^2} dx =$$

$$\left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a$$

$$= \frac{a^2}{2} \sin^{-1}(1) = \frac{\pi a^2}{4}$$

$$2) \int_0^a (a - x) dx = \left[ax - \frac{x^2}{2} \right]_0^a = \frac{a^2}{2}$$

$$A = \frac{b}{a} \left(\frac{\pi a^2}{4} - \frac{a^2}{2} \right)$$

$$A = \frac{b}{a} \cdot a^2 \left(\frac{\pi}{4} - \frac{1}{2} \right) = ab \left(\frac{\pi - 2}{4} \right)$$